

MLM — Mutation Logging Module

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Category: Governance & Enforcement

Module: Mutation Logging Module (MLM)

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Standard: Self-Evolving Systems Control Standard (SESCS)

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Status: Canonical Module

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Authority: OOF™ Origin Open Foundation™

Protection: MIP™ — Methodological Intellectual Property

Canonical Language: English (UCL™)

Compatibility:

SESCS · MTVF · TSA · EVIP · ArtData · AI Governance Systems · ORGS

Canonical Definition System

Mutation Logging Module (MLM) defines the mandatory recording, traceability, and audit structure for all system mutations occurring within self-evolving systems.

Base

Without mutation logging:

- system changes become invisible
- errors cannot be traced
- responsibility cannot be assigned
- audit is impossible

A system that cannot explain its changes cannot be trusted.

Protocol

Any system capable of mutation **MUST**:

- record every mutation event
- store mutation data in immutable form
- assign timestamp and origin to each change
- ensure auditability of mutation history

Absence of mutation logging results in:

Non-Traceable System State (NTS)

Scope

MLM applies to:

- AI agents generating new logic
- systems modifying execution pathways
- autonomous optimization systems
- adaptive enterprise systems
- runtime mutation environments

Methodology

1. detect mutation event
2. capture mutation data
3. assign timestamp
4. identify mutation origin
5. store mutation record
6. enable audit access

Mutation Record Structure

Each mutation MUST include:

- mutation identifier
- timestamp
- origin (human / system / hybrid)
- affected component
- description of change
- pre-state reference
- post-state reference

Structural Rule (Critical)

No mutation is valid unless it is recorded.

Immutability Requirement

Mutation logs MUST be:

- tamper-resistant
- append-only
- verifiable

Deletion or alteration of logs is prohibited.

MIF — Minimal Implementation Framework

Step 1 — Mutation Detection

Detect any system change.

Minimum:

- mutation trigger captured
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Step 2 — Record Creation

Create mutation record.

Minimum:

- timestamp
 - mutation identifier
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Step 3 — Origin Identification

Assign origin.

Minimum:

- human / system / hybrid
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Step 4 — Storage

Store mutation log.

Minimum:

- persistent storage
 - non-editable format
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MIF+ — Extended Implementation Framework

Includes all MIF requirements plus:

- cryptographic log verification
 - distributed logging systems
 - real-time logging streams
 - cross-system log synchronization
 - integration with TSA classification
 - anomaly detection in mutation logs
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Decision Logic

Mutation state is:

TRACEABLE → recorded and verifiable

NON-TRACEABLE → missing or incomplete logging

Use Case 1 — AI System Behavior Change

Scenario

AI system modifies its decision logic.

Problem

- change is not recorded
- behavior becomes unpredictable
- no traceability

Application

MLM enforces:

- full mutation logging
- timestamped change tracking

Result

- system behavior is traceable
 - changes can be audited
 - trust is maintained
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Use Case 2 — System Failure Investigation

Scenario

System produces incorrect or harmful output.

Problem

- no record of system changes
- failure source cannot be identified

Application

MLM provides:

- full mutation history
- traceable change sequence

Result

- root cause identified
 - responsibility assigned
 - system corrected
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Compatibility

MLM operates with:

- MCM — defines mutation boundaries
 - MVM — validates mutations
 - RAM — assigns responsibility
 - RIM — enables rollback
 - TSA — classifies trust
 - MTVF — validation layer
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Structural Principle

If a system changes, the change must be recorded.

Closing Statement

Unrecorded change is invisible risk.

Invisible risk cannot be controlled.

Mutation Logging Module (MLM) establishes a non-negotiable condition:

Every mutation must be recorded, traceable, and auditable.

Related Documents

[→ SESCO Standard](#)

[→ SESCO — Master Index](#)

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Canonical Source

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